Reg. No. [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] **[18CE731]**

SEVENTH SEMESTER B.TECH EXAMINATION, DECEMBER-2021

DESIGN AND DRAWING OF STEEL STRUCTURES

Time: 3 Hours **Maximum Marks: 70**

Instructions: i. PART – A, Question number 1 and 2 are compulsory. Each question carries 14 marks.

ii. PART – B, Answer atleast one full question. Each question carries 42 marks.

iii. Use Drawing sheets to draw the diagrams. All drawing should be done by the aid of drafter/Tee Square.

iv: use of IS 800-2007 and Steel Tables are Permitted.

v: Missing data may be suitably assumed.

PART- A

1. Two secondary beams ISLB-300@37.7kg/m and ISMB-500 are connected to main beam ISMB-600. Size of cleat angle ISA 90x90x8mm. **14**

Provide

3-#20bolts for angle ISLB-300 connecting with ISMB 600

4-#20 bolts for angle ISMB-500 connecting with ISMB 600

Assume Top Surface of beams are in same level.

Draw to a suitable scale. i) Front view ii) Any one side view

2. A Column splice made between to ISMB 400@61.6 kg/m. The splicing is made by using web plate of size 320mm x 300mm x 8mm along with two flange plates of size 400x100x16mm. Flange plates and web strap plates are welded to ISMB 400 by 6mm and 8mm continuous fillet weld. **14**

Draw to a suitable scale

i. Front view.

ii. Side view.

PART- B

3. Design a Gantry girder to an industrial shed to support an electric overhead travelling crane, using the following data: **42**

Crane capacity = 250kN

Weight of Crab (trolley) = 60kN

Span of crane girder between rails = 18m

Weight of crane girder (excluding trolley) = 200kN

Minimum approach for crane hook = 1.2m

Wheel base = 3.2m

Span of gantry girder = 7m

Mass of rail section = 300 N/m (0.3kN/m)

Height of rail section = 75mm

Yield stress f_y = 250 N/mm^2

Draw to a suitable scale the following

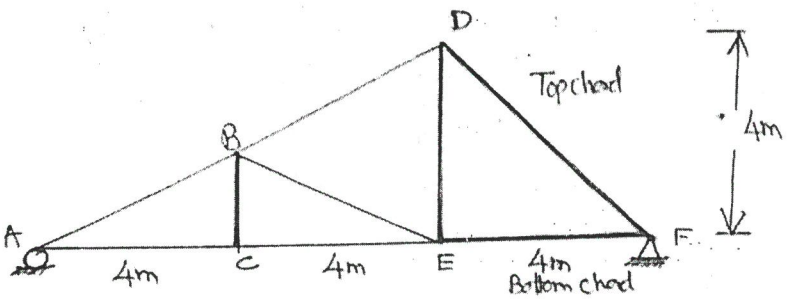
i. Front view

ii. Top view

iii. Section of gantry girder to an enlarged scale.

OR

4. Design the members FD, FE, DE and BC of the truss shown in figure. Use bolted or welded connection. Also design bearing plate if support reaction is 120kN. **42**



Sl. no.	Member	Force (kN)	Length (m)
1	FD	100kN (C)	5.66m
2	FE	80kN (T)	4m
3	DE	80kN (T)	4m
4	BC	50kN (C)	2m